

# Local + BRV-paper Results — Baseline vs. Uniform $\lambda=0.9$

Each row compares one benchmark across two methods (Baseline / +Uniform 0.9) and two sources (Local re-scored eval / BRV Chen et al. 2025 Tables 2-3). **Gap** = **Local** – **Paper**; cells where |Gap| > 1 pp are highlighted in red. The  $\Delta$  columns are also kept — Local  $\Delta$  gets the original yellow/orange/pink/red coloring (favorable/perception/reasoning/unfavorable).

## LLaVA-Next-LLaMA3-8B (task vector: Dart-Prop)

Local re-scored vs. BRV Table 2 (+Dart-Prop column). MathVerse-Overall is computed locally as the simple mean of the five mode scores, matching BRV's methodology (the five modes share the same 788 problems).

| Benchmark $\uparrow$    | Type       | Baseline |       |       | +Uniform 0.9 |       |       | $\Delta$ |       |
|-------------------------|------------|----------|-------|-------|--------------|-------|-------|----------|-------|
|                         |            | Local    | Paper | Gap   | Local        | Paper | Gap   | Local    | Paper |
| MathVista-Overall       | perception | 37.80    | 37.40 | +0.40 | 37.90        | 38.00 | -0.10 | +0.10    | +0.60 |
| MathVista-General (NEW) | perception | 51.96    | 51.70 | +0.26 | 49.13        | 48.70 | +0.43 | -2.83    | -3.00 |
| MathVista-Math (NEW)    | reasoning  | 25.74    | 25.40 | +0.34 | 28.33        | 28.90 | -0.57 | +2.59    | +3.50 |
| MathVerse-Overall       | reasoning  | 18.45    | 20.10 | -1.65 | 19.03        | 23.70 | -4.67 | +0.58    | +3.60 |
| MathVerse-T-D           | reasoning  | 22.34    | 25.90 | -3.56 | 23.98        | 30.70 | -6.72 | +1.64    | +4.80 |
| MathVerse-T-L           | reasoning  | 19.67    | 20.80 | -1.13 | 19.92        | 24.80 | -4.88 | +0.25    | +4.00 |
| MathVerse-V-I           | reasoning  | 19.80    | 21.10 | -1.30 | 20.18        | 25.50 | -5.32 | +0.38    | +4.40 |
| MathVerse-V-D           | reasoning  | 15.48    | 16.50 | -1.02 | 15.48        | 19.80 | -4.32 | +0.00    | +3.30 |
| MathVerse-V-O           | reasoning  | 14.97    | 16.00 | -1.03 | 15.61        | 17.40 | -1.79 | +0.64    | +1.40 |
| MathVision              | reasoning  | 14.14    | 13.80 | +0.34 | 14.47        | 14.80 | -0.33 | +0.33    | +1.00 |
| MMStar-Overall          | perception | 43.33    | 43.80 | -0.47 | 43.60        | 43.60 | +0.00 | +0.27    | -0.20 |
| MMStar-Math             | reasoning  | 29.60    | 30.00 | -0.40 | 33.60        | 33.60 | +0.00 | +4.00    | +3.60 |
| DynaMath-Average        | reasoning  | 22.75    | 22.70 | +0.05 | 24.35        | 24.50 | -0.15 | +1.60    | +1.80 |
| DynaMath-Worst (NEW)    | reasoning  | 3.19     | —     | —     | 4.39         | —     | —     | +1.20    | —     |

Idefics2-8B (task vector: MAMmoTH-1)

Local re-scored vs. BRV Table 3 (+MAMmoTH-1 column).

| Benchmark ↑             | Type       | Baseline |       |       | +Uniform 0.9 |       |       | Δ     |       |
|-------------------------|------------|----------|-------|-------|--------------|-------|-------|-------|-------|
|                         |            | Local    | Paper | Gap   | Local        | Paper | Gap   | Local | Paper |
| MathVista-Overall       | perception | 52.00    | 51.80 | +0.20 | 52.50        | 53.00 | -0.50 | +0.50 | +1.20 |
| MathVista-General (NEW) | perception | 56.30    | 57.00 | -0.70 | 59.13        | 58.50 | +0.63 | +2.83 | +1.50 |
| MathVista-Math (NEW)    | reasoning  | 48.33    | 47.40 | +0.93 | 46.85        | 48.30 | -1.45 | -1.48 | +0.90 |
| MathVerse-Overall       | reasoning  | 18.38    | 19.40 | -1.02 | 18.96        | 20.40 | -1.44 | +0.58 | +1.00 |
| MathVerse-T-D           | reasoning  | 21.70    | 24.40 | -2.70 | 24.62        | 26.00 | -1.38 | +2.92 | +1.60 |
| MathVerse-T-L           | reasoning  | 20.81    | 21.30 | -0.49 | 21.57        | 22.50 | -0.93 | +0.76 | +1.20 |
| MathVerse-V-I           | reasoning  | 20.05    | 20.70 | -0.65 | 19.67        | 21.30 | -1.63 | -0.38 | +0.60 |
| MathVerse-V-D           | reasoning  | 19.04    | 19.70 | -0.66 | 17.89        | 19.80 | -1.91 | -1.15 | +0.10 |
| MathVerse-V-O           | reasoning  | 10.28    | 11.00 | -0.72 | 11.04        | 12.10 | -1.06 | +0.76 | +1.10 |
| MathVision              | reasoning  | 16.45    | 17.10 | -0.65 | 14.80        | 16.80 | -2.00 | -1.65 | -0.30 |
| MMStar-Overall          | perception | 49.33    | 49.50 | -0.17 | 48.40        | 48.30 | +0.10 | -0.93 | -1.20 |
| MMStar-Math             | reasoning  | 39.60    | 39.60 | +0.00 | 40.00        | 40.80 | -0.80 | +0.40 | +1.20 |
| DynaMath-Average        | reasoning  | 21.80    | 21.80 | +0.00 | 23.01        | 23.20 | -0.19 | +1.21 | +1.40 |
| DynaMath-Worst (NEW)    | reasoning  | 3.19     | —     | —     | 4.79         | —     | —     | +1.60 | —     |

Qwen2-VL-7B-Instruct (task vector: Qwen2-Math-7B)

Local re-scored vs. BRV Appendix E / Table 7 (+Qwen2-Math column).

| Benchmark ↑             | Type       | Baseline |       |       | +Uniform 0.9 |       |       | Δ     |       |
|-------------------------|------------|----------|-------|-------|--------------|-------|-------|-------|-------|
|                         |            | Local    | Paper | Gap   | Local        | Paper | Gap   | Local | Paper |
| MathVista-Overall       | perception | 61.10    | 61.20 | -0.10 | 58.30        | 60.20 | -1.90 | -2.80 | -1.00 |
| MathVista-General (NEW) | perception | 69.78    | 69.60 | +0.18 | 66.09        | 68.00 | -1.91 | -3.69 | -1.60 |
| MathVista-Math (NEW)    | reasoning  | 53.70    | 54.10 | -0.40 | 51.67        | 53.50 | -1.83 | -2.03 | -0.60 |
| MathVerse-Overall       | reasoning  | 30.68    | 31.80 | -1.12 | 30.41        | 31.90 | -1.49 | -0.27 | +0.10 |
| MathVerse-T-D           | reasoning  | 34.77    | 35.90 | -1.13 | 33.88        | 37.10 | -3.22 | -0.89 | +1.20 |
| MathVerse-T-L           | reasoning  | 30.58    | 31.40 | -0.82 | 30.08        | 31.70 | -1.62 | -0.50 | +0.30 |
| MathVerse-V-I           | reasoning  | 30.71    | 31.50 | -0.79 | 30.08        | 31.50 | -1.42 | -0.63 | +0.00 |
| MathVerse-V-D           | reasoning  | 32.23    | 33.10 | -0.87 | 31.09        | 32.50 | -1.41 | -1.14 | -0.60 |
| MathVerse-V-O           | reasoning  | 25.13    | 26.90 | -1.77 | 26.90        | 26.70 | +0.20 | +1.77 | -0.20 |
| MathVision              | reasoning  | 20.72    | 21.10 | -0.38 | 20.07        | 21.70 | -1.63 | -0.65 | +0.60 |
| MMStar-Overall          | perception | 59.60    | 59.90 | -0.30 | 56.47        | 59.50 | -3.03 | -3.13 | -0.40 |
| MMStar-Math             | reasoning  | 58.40    | 59.20 | -0.80 | 55.20        | 58.40 | -3.20 | -3.20 | -0.80 |
| DynaMath-Average        | reasoning  | 33.55    | 34.40 | -0.85 | 31.40        | 35.00 | -3.60 | -2.15 | +0.60 |
| DynaMath-Worst (NEW)    | reasoning  | 10.38    | —     | —     | 10.78        | —     | —     | +0.40 | —     |

LLaVA-1.5-7B (task vector: MetaMath-7B-V1.0)

Local re-scored only. BRV Tables 2/3/7 do not include LLaVA-1.5-7B. MMStar baseline + uniform jobs in flight as of 2026-06-04. POPE-Overall available (84.80 / 84.10,  $\Delta -0.70$ ) — see supplement notes.

| Benchmark $\uparrow$    | Type       | Baseline |       |     | +Uniform 0.9 |       |     | $\Delta$ |       |
|-------------------------|------------|----------|-------|-----|--------------|-------|-----|----------|-------|
|                         |            | Local    | Paper | Gap | Local        | Paper | Gap | Local    | Paper |
| MathVista-Overall       | perception | 24.10    | —     | —   | 23.70        | —     | —   | -0.40    | —     |
| MathVista-General (NEW) | perception | 24.21    | —     | —   | 23.63        | —     | —   | -0.58    | —     |
| MathVista-Math (NEW)    | reasoning  | 17.04    | —     | —   | 18.33        | —     | —   | +1.29    | —     |
| MathVerse-Overall       | reasoning  | 15.03    | —     | —   | 14.90        | —     | —   | -0.13    | —     |
| MathVerse-T-D           | reasoning  | 15.61    | —     | —   | 14.85        | —     | —   | -0.76    | —     |
| MathVerse-T-L           | reasoning  | 14.59    | —     | —   | 14.59        | —     | —   | +0.00    | —     |
| MathVerse-V-I           | reasoning  | 14.85    | —     | —   | 14.72        | —     | —   | -0.13    | —     |
| MathVerse-V-D           | reasoning  | 15.36    | —     | —   | 15.10        | —     | —   | -0.26    | —     |
| MathVerse-V-O           | reasoning  | 14.72    | —     | —   | 15.23        | —     | —   | +0.51    | —     |
| MathVision              | reasoning  | 15.79    | —     | —   | 17.11        | —     | —   | +1.32    | —     |
| MMStar-Overall          | perception | 31.07    | —     | —   | —            | —     | —   | —        | —     |
| MMStar-Math             | reasoning  | 22.40    | —     | —   | —            | —     | —   | —        | —     |
| DynaMath-Average        | reasoning  | 12.73    | —     | —   | 11.86        | —     | —   | -0.87    | —     |
| DynaMath-Worst (NEW)    | reasoning  | 1.20     | —     | —   | 1.80         | —     | —   | +0.60    | —     |

LLaVA-Next-Mistral-7B (task vector: MAMmoTH-7B-Mistral)

Local re-scored only. BRV Tables 2/3/7 do not include LLaVA-Next-Mistral-7B. MM-Math jobs in flight as of 2026-06-04 (~8–12h remaining; uniform at 76%, baseline at 68%). Bench rows not in this table but in the supplement: MME 1788.42 / 1791.94, HallusionBench 30.28 / 33.44, POPE-Overall 87.44 / 87.36 ( $\Delta -0.08$ ).

| Benchmark $\uparrow$    | Type       | Baseline |       |     | +Uniform 0.9 |       |     | $\Delta$ |       |
|-------------------------|------------|----------|-------|-----|--------------|-------|-----|----------|-------|
|                         |            | Local    | Paper | Gap | Local        | Paper | Gap | Local    | Paper |
| MathVista-Overall       | perception | 35.40    | —     | —   | 36.00        | —     | —   | +0.60    | —     |
| MathVista-General (NEW) | perception | 35.96    | —     | —   | 36.74        | —     | —   | +0.78    | —     |
| MathVista-Math (NEW)    | reasoning  | 26.11    | —     | —   | 26.30        | —     | —   | +0.19    | —     |
| MathVerse-Overall       | reasoning  | 18.27    | —     | —   | 17.92        | —     | —   | -0.35    | —     |
| MathVerse-T-D           | reasoning  | 21.19    | —     | —   | 21.07        | —     | —   | -0.12    | —     |
| MathVerse-T-L           | reasoning  | 17.89    | —     | —   | 18.27        | —     | —   | +0.38    | —     |
| MathVerse-V-I           | reasoning  | 19.16    | —     | —   | 19.16        | —     | —   | +0.00    | —     |
| MathVerse-V-D           | reasoning  | 17.01    | —     | —   | 16.75        | —     | —   | -0.26    | —     |
| MathVerse-V-O           | reasoning  | 16.12    | —     | —   | 14.34        | —     | —   | -1.78    | —     |
| MathVision              | reasoning  | 11.18    | —     | —   | 13.82        | —     | —   | +2.64    | —     |
| MMStar-Overall          | perception | 39.33    | —     | —   | —            | —     | —   | —        | —     |
| MMStar-Math             | reasoning  | 27.20    | —     | —   | —            | —     | —   | —        | —     |
| DynaMath-Average        | reasoning  | 17.78    | —     | —   | 19.26        | —     | —   | +1.48    | —     |
| DynaMath-Worst (NEW)    | reasoning  | 2.20     | —     | —   | 2.40         | —     | —   | +0.20    | —     |

## All values, row-by-row (consolidated reference)

**Coverage:** 8 of the 9 enrichment-target VLMs now appear in this table (LLaVA-1.5-7B, Llafics2-8B, LLaVA-Next-LLaMA3-8B, LLaVA-Next-Mistral-7B, Qwen2-VL-7B-Instruct, LLaVA-OneVision-7B, InternVL2.5-8B, Qwen2.5-VL-7B). The 3 newly-added models have partial cells (uniform-only or baseline-only) because their eval jobs are still in flight or only one side has landed so far. Qwen2.5-VL-3B remains structurally blocked (no shape-compatible math donor); see the “Blocked models” page for the full blocker map.

**Note on (NEW) rows for LLaVA-1.5 and Mistral:** the MathVista-General, MathVista-Math, and DynaMath-Worst values for those two pairs are computed by `extract_merge_coverage.py` from per-skill MathVista CSV rows / row 3 of the DynaMath CSV. MathVista-Math and DynaMath-Worst are exact matches against the canonical extraction. MathVista-General is approximated as (Overall – Math) weighted by row count; BRV's General excludes additional math-leaning skills (arithmetic, algebra, geometry, etc.) so its published value tends to run ~5 pp higher than this approximation. The 3 existing models (LLaVA-LLaMA3, Llafics2, Qwen2-VL) keep the original BRV-script values.

**Hook-aware columns:** the three rightmost columns show the best (max-scoring) value across **every  $\alpha$ -triple variant on disk** for each (model, benchmark) cell. PMBT-gate scans all `pmbt_t*_hgate` mode-tags (~20 variants); PMBT-gate\_up scans all `pmbt_*` tags without the `_hgate` suffix (~206 variants); FT-gate scans all `ft_pmbt*_hgate` tags (~10 variants). MathVerse-Overall is the mean of the 5 split rows for the best single mode-tag per cell (not a free max across modes). Cells = — indicate that no on-disk variant in that hook+method group has a value for that (model, tv, benchmark) tuple. PMBT-gate and PMBT-gate\_up are not perfectly comparable: the on-disk gate variants don't carry the `_a1.0_o1.0` suffix that gate\_up variants do, so  $\alpha_{\text{other}}$  differs (gate uses default 0.9; gate\_up uses 1.0).

Same data as the per-model tables above, flattened into one table for spreadsheet-style scanning. One row per (model × benchmark). **n (Local)** is the row count we measure in the local ~/LMUData/ TSV; **n (BRV)** is the row count BRV's eval registers via `VLMEvalKit_br/vlmeval/dataset/image_vqa.py`. The two columns are identical for every row because BRV's dataset registry resolves to the same upstream files (MD5-matched). The asterisk on MathVerse-Overall (788\*) marks the *effective* n — the five modes share the same 788 problems, so the 5-mode average denominator stays at 788, not 3,940. Red highlight on Gap cells = |Local – Paper| > 1 pp.

| Model                            | Benchmark               | n (Local) | n (BRV) | Type       | Baseline |       |       | +Uniform 0.9 |       |       | $\Delta$ |       | Hook-aware best (max across $\alpha$ -triples) |              |         |
|----------------------------------|-------------------------|-----------|---------|------------|----------|-------|-------|--------------|-------|-------|----------|-------|------------------------------------------------|--------------|---------|
|                                  |                         |           |         |            | Local    | Paper | Gap   | Local        | Paper | Gap   | Local    | Paper | PMBT gate                                      | PMBT gate_up | FT gate |
| LLaVA-Next-LLaMA3-8B (Dart-Prop) | MathVista-Overall       | 1000      | 1000    | perception | 37.80    | 37.40 | +0.40 | 37.90        | 38.00 | -0.10 | +0.10    | +0.60 | 38.10                                          | 39.50        | —       |
|                                  | MathVista-General (NEW) | 460       | 460     | perception | 51.96    | 51.70 | +0.26 | 49.13        | 48.70 | +0.43 | -2.83    | -3.00 | 38.71                                          | 40.38        | —       |
|                                  | MathVista-Math (NEW)    | 540       | 540     | reasoning  | 25.74    | 25.40 | +0.34 | 28.33        | 28.90 | -0.57 | +2.59    | +3.50 | 27.96                                          | 31.48        | —       |
|                                  | MathVerse-Overall       | 788*      | 788*    | reasoning  | 18.45    | 20.10 | -1.65 | 19.03        | 23.70 | -4.67 | +0.58    | +3.60 | 19.70                                          | 20.33        | —       |
|                                  | MathVerse-T-D           | 788       | 788     | reasoning  | 22.34    | 25.90 | -3.56 | 23.98        | 30.70 | -6.72 | +1.64    | +4.80 | 25.51                                          | 27.16        | —       |
|                                  | MathVerse-T-L           | 788       | 788     | reasoning  | 19.67    | 20.80 | -1.13 | 19.92        | 24.80 | -4.88 | +0.25    | +4.00 | 21.45                                          | 22.34        | —       |
|                                  | MathVerse-V-I           | 788       | 788     | reasoning  | 19.80    | 21.10 | -1.30 | 20.18        | 25.50 | -5.32 | +0.38    | +4.40 | 21.57                                          | 22.34        | —       |
|                                  | MathVerse-V-D           | 788       | 788     | reasoning  | 15.48    | 16.50 | -1.02 | 15.48        | 19.80 | -4.32 | +0.00    | +3.30 | 16.50                                          | 17.64        | —       |
|                                  | MathVerse-V-O           | 788       | 788     | reasoning  | 14.97    | 16.00 | -1.03 | 15.61        | 17.40 | -1.79 | +0.64    | +1.40 | 15.36                                          | 16.62        | —       |
|                                  | MathVision              | 304       | 304     | reasoning  | 14.14    | 13.80 | +0.34 | 14.47        | 14.80 | -0.33 | +0.33    | +1.00 | 14.80                                          | 18.75        | —       |
|                                  | MMStar-Overall          | 1500      | 1500    | perception | 43.33    | 43.80 | -0.47 | 43.60        | 43.60 | +0.00 | +0.27    | -0.20 | 43.93                                          | 44.07        | 41.67   |
|                                  | MMStar-Math             | 250       | 250     | reasoning  | 29.60    | 30.00 | -0.40 | 33.60        | 33.60 | +0.00 | +4.00    | +3.60 | 32.80                                          | 35.20        | 32.40   |

|                                      |                         |           |       |            | Baseline |       |       | +Uniform 0.9 |       |       | $\Delta$ |       | Hook-aware best (max across $\alpha$ -triples) |              |         |
|--------------------------------------|-------------------------|-----------|-------|------------|----------|-------|-------|--------------|-------|-------|----------|-------|------------------------------------------------|--------------|---------|
| Model                                | Benchmark               | n (Local) | (BRV) | Type       | Local    | Paper | Gap   | Local        | Paper | Gap   | Local    | Paper | PMBT gate                                      | PMBT gate_up | FT gate |
|                                      | DynaMath-Average        | 5010      | 5010  | reasoning  | 22.75    | 22.70 | +0.05 | 24.35        | 24.50 | -0.15 | +1.60    | +1.80 | 24.53                                          | 25.25        | —       |
|                                      | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 3.19     | —     | —     | 4.39         | —     | —     | +1.20    | —     | 4.19                                           | 5.39         | —       |
| Idefics2-8B (MAmmoTH-1)              | MathVista-Overall       | 1000      | 1000  | perception | 52.00    | 51.80 | +0.20 | 52.50        | 53.00 | -0.50 | +0.50    | +1.20 | 53.20                                          | 53.60        | 52.90   |
|                                      | MathVista-General (NEW) | 460       | 460   | perception | 56.30    | 57.00 | -0.70 | 59.13        | 58.50 | +0.63 | +2.83    | +1.50 | 52.06                                          | 52.74        | 51.99   |
|                                      | MathVista-Math (NEW)    | 540       | 540   | reasoning  | 48.33    | 47.40 | +0.93 | 46.85        | 48.30 | -1.45 | -1.48    | +0.90 | 48.70                                          | 49.26        | 48.89   |
|                                      | MathVerse-Overall       | 788*      | 788*  | reasoning  | 18.38    | 19.40 | -1.02 | 18.96        | 20.40 | -1.44 | +0.58    | +1.00 | 19.62                                          | 19.39        | 19.57   |
|                                      | MathVerse-T-D           | 788       | 788   | reasoning  | 21.70    | 24.40 | -2.70 | 24.62        | 26.00 | -1.38 | +2.92    | +1.60 | 23.48                                          | 23.98        | 23.48   |
|                                      | MathVerse-T-L           | 788       | 788   | reasoning  | 20.81    | 21.30 | -0.49 | 21.57        | 22.50 | -0.93 | +0.76    | +1.20 | 21.70                                          | 21.83        | 21.07   |
|                                      | MathVerse-V-I           | 788       | 788   | reasoning  | 20.05    | 20.70 | -0.65 | 19.67        | 21.30 | -1.63 | -0.38    | +0.60 | 21.70                                          | 21.70        | 20.69   |
|                                      | MathVerse-V-D           | 788       | 788   | reasoning  | 19.04    | 19.70 | -0.66 | 17.89        | 19.80 | -1.91 | -1.15    | +0.10 | 19.67                                          | 19.67        | 18.91   |
|                                      | MathVerse-V-O           | 788       | 788   | reasoning  | 10.28    | 11.00 | -0.72 | 11.04        | 12.10 | -1.06 | +0.76    | +1.10 | 12.94                                          | 13.32        | 14.72   |
|                                      | MathVision              | 304       | 304   | reasoning  | 16.45    | 17.10 | -0.65 | 14.80        | 16.80 | -2.00 | -1.65    | -0.30 | 18.42                                          | 18.09        | 17.43   |
|                                      | MMStar-Overall          | 1500      | 1500  | perception | 49.33    | 49.50 | -0.17 | 48.40        | 48.30 | +0.10 | -0.93    | -1.20 | 49.27                                          | 49.60        | 49.47   |
|                                      | MMStar-Math             | 250       | 250   | reasoning  | 39.60    | 39.60 | +0.00 | 40.00        | 40.80 | -0.80 | +0.40    | +1.20 | 42.00                                          | 43.20        | 41.60   |
|                                      | DynaMath-Average        | 5010      | 5010  | reasoning  | 21.80    | 21.80 | +0.00 | 23.01        | 23.20 | -0.19 | +1.21    | +1.40 | 23.87                                          | 24.07        | 23.19   |
|                                      | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 3.19     | —     | —     | 4.79         | —     | —     | +1.60    | —     | 5.19                                           | 5.79         | 5.19    |
| Qwen2-VL-7B-Instruct (Qwen2-Math-7B) | MathVista-Overall       | 1000      | 1000  | perception | 61.10    | 61.20 | -0.10 | 58.30        | 60.20 | -1.90 | -2.80    | -1.00 | 61.10                                          | 61.10        | —       |
|                                      | MathVista-General (NEW) | 460       | 460   | perception | 69.78    | 69.60 | +0.18 | 66.09        | 68.00 | -1.91 | -3.69    | -1.60 | 60.33                                          | 60.47        | —       |
|                                      | MathVista-Math (NEW)    | 540       | 540   | reasoning  | 53.70    | 54.10 | -0.40 | 51.67        | 53.50 | -1.83 | -2.03    | -0.60 | 53.89                                          | 53.52        | —       |
|                                      | MathVerse-Overall       | 788*      | 788*  | reasoning  | 30.68    | 31.80 | -1.12 | 30.41        | 31.90 | -1.49 | -0.27    | +0.10 | 30.99                                          | 30.71        | —       |
|                                      | MathVerse-T-D           | 788       | 788   | reasoning  | 34.77    | 35.90 | -1.13 | 33.88        | 37.10 | -3.22 | -0.89    | +1.20 | 34.90                                          | 34.52        | —       |
|                                      | MathVerse-T-L           | 788       | 788   | reasoning  | 30.58    | 31.40 | -0.82 | 30.08        | 31.70 | -1.62 | -0.50    | +0.30 | 31.09                                          | 30.96        | —       |
|                                      | MathVerse-V-I           | 788       | 788   | reasoning  | 30.71    | 31.50 | -0.79 | 30.08        | 31.50 | -1.42 | -0.63    | +0.00 | 30.96                                          | 31.35        | —       |
|                                      | MathVerse-V-D           | 788       | 788   | reasoning  | 32.23    | 33.10 | -0.87 | 31.09        | 32.50 | -1.41 | -1.14    | -0.60 | 32.61                                          | 31.73        | —       |

|                                            |                         |           |       |            | Baseline |       |       | +Uniform 0.9 |       |       | $\Delta$ |       | Hook-aware best (max across $\alpha$ -triples) |              |         |
|--------------------------------------------|-------------------------|-----------|-------|------------|----------|-------|-------|--------------|-------|-------|----------|-------|------------------------------------------------|--------------|---------|
| Model                                      | Benchmark               | n (Local) | (BRV) | Type       | Local    | Paper | Gap   | Local        | Paper | Gap   | Local    | Paper | PMBT gate                                      | PMBT gate_up | FT gate |
|                                            | MathVerse-V-O           | 788       | 788   | reasoning  | 25.13    | 26.90 | -1.77 | 26.90        | 26.70 | +0.20 | +1.77    | -0.20 | 27.16                                          | 26.65        | —       |
|                                            | MathVision              | 304       | 304   | reasoning  | 20.72    | 21.10 | -0.38 | 20.07        | 21.70 | -1.63 | -0.65    | +0.60 | 20.72                                          | 21.71        | —       |
|                                            | MMStar-Overall          | 1500      | 1500  | perception | 59.60    | 59.90 | -0.30 | 56.47        | 59.50 | -3.03 | -3.13    | -0.40 | 59.80                                          | 59.60        | —       |
|                                            | MMStar-Math             | 250       | 250   | reasoning  | 58.40    | 59.20 | -0.80 | 55.20        | 58.40 | -3.20 | -3.20    | -0.80 | 58.00                                          | 59.20        | —       |
|                                            | DynaMath-Average        | 5010      | 5010  | reasoning  | 33.55    | 34.40 | -0.85 | 31.40        | 35.00 | -3.60 | -2.15    | +0.60 | 33.57                                          | 33.73        | —       |
|                                            | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 10.38    | —     | —     | 10.78        | —     | —     | +0.40    | —     | 10.38                                          | 10.58        | —       |
| LLaVA-1.5-7B (MetaMath-7B-V1.0)            | MathVista-Overall       | 1000      | 1000  | perception | 24.10    | —     | —     | 23.70        | —     | —     | -0.40    | —     | —                                              | —            | —       |
|                                            | MathVista-General (NEW) | 460       | 460   | perception | 24.21    | —     | —     | 23.63        | —     | —     | -0.58    | —     | —                                              | —            | —       |
|                                            | MathVista-Math (NEW)    | 540       | 540   | reasoning  | 17.04    | —     | —     | 18.33        | —     | —     | +1.29    | —     | —                                              | —            | —       |
|                                            | MathVerse-Overall       | 788*      | 788*  | reasoning  | 15.03    | —     | —     | 14.90        | —     | —     | -0.13    | —     | —                                              | —            | —       |
|                                            | MathVerse-T-D           | 788       | 788   | reasoning  | 15.61    | —     | —     | 14.85        | —     | —     | -0.76    | —     | —                                              | —            | —       |
|                                            | MathVerse-T-L           | 788       | 788   | reasoning  | 14.59    | —     | —     | 14.59        | —     | —     | +0.00    | —     | —                                              | —            | —       |
|                                            | MathVerse-V-I           | 788       | 788   | reasoning  | 14.85    | —     | —     | 14.72        | —     | —     | -0.13    | —     | —                                              | —            | —       |
|                                            | MathVerse-V-D           | 788       | 788   | reasoning  | 15.36    | —     | —     | 15.10        | —     | —     | -0.26    | —     | —                                              | —            | —       |
|                                            | MathVerse-V-O           | 788       | 788   | reasoning  | 14.72    | —     | —     | 15.23        | —     | —     | +0.51    | —     | —                                              | —            | —       |
|                                            | MathVision              | 304       | 304   | reasoning  | 15.79    | —     | —     | 17.11        | —     | —     | +1.32    | —     | —                                              | —            | —       |
|                                            | MMStar-Overall          | 1500      | 1500  | perception | 31.07    | —     | —     | —            | —     | —     | —        | —     | —                                              | —            | —       |
|                                            | MMStar-Math             | 250       | 250   | reasoning  | 22.40    | —     | —     | —            | —     | —     | —        | —     | —                                              | —            | —       |
|                                            | DynaMath-Average        | 5010      | 5010  | reasoning  | 12.73    | —     | —     | 11.86        | —     | —     | -0.87    | —     | —                                              | —            | —       |
|                                            | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 1.20     | —     | —     | 1.80         | —     | —     | +0.60    | —     | —                                              | —            | —       |
| LLaVA-Next-Mistral-7B (MAMmoTH-7B-Mistral) | MathVista-Overall       | 1000      | 1000  | perception | 35.40    | —     | —     | 36.00        | —     | —     | +0.60    | —     | 34.80                                          | —            | —       |
|                                            | MathVista-General (NEW) | 460       | 460   | perception | 35.96    | —     | —     | 36.74        | —     | —     | +0.78    | —     | 35.48                                          | —            | —       |
|                                            | MathVista-Math (NEW)    | 540       | 540   | reasoning  | 26.11    | —     | —     | 26.30        | —     | —     | +0.19    | —     | 24.81                                          | —            | —       |



|                                    |                         |           |       |            | Baseline |       |     | +Uniform 0.9 |       |     | $\Delta$ |       | Hook-aware best (max across $\alpha$ -triples) |              |         |
|------------------------------------|-------------------------|-----------|-------|------------|----------|-------|-----|--------------|-------|-----|----------|-------|------------------------------------------------|--------------|---------|
| Model                              | Benchmark               | n (Local) | (BRV) | Type       | Local    | Paper | Gap | Local        | Paper | Gap | Local    | Paper | PMBT gate                                      | PMBT gate_up | FT gate |
|                                    | MathVerse-Overall       | 788*      | 788*  | reasoning  | 18.27    | —     | —   | 17.92        | —     | —   | -0.35    | —     | 18.30                                          | —            | —       |
|                                    | MathVerse-T-D           | 788       | 788   | reasoning  | 21.19    | —     | —   | 21.07        | —     | —   | -0.12    | —     | 21.07                                          | —            | —       |
|                                    | MathVerse-T-L           | 788       | 788   | reasoning  | 17.89    | —     | —   | 18.27        | —     | —   | +0.38    | —     | 18.91                                          | —            | —       |
|                                    | MathVerse-V-I           | 788       | 788   | reasoning  | 19.16    | —     | —   | 19.16        | —     | —   | +0.00    | —     | 19.67                                          | —            | —       |
|                                    | MathVerse-V-D           | 788       | 788   | reasoning  | 17.01    | —     | —   | 16.75        | —     | —   | -0.26    | —     | 16.88                                          | —            | —       |
|                                    | MathVerse-V-O           | 788       | 788   | reasoning  | 16.12    | —     | —   | 14.34        | —     | —   | -1.78    | —     | 14.97                                          | —            | —       |
|                                    | MathVision              | 304       | 304   | reasoning  | 11.18    | —     | —   | 13.82        | —     | —   | +2.64    | —     | 13.16                                          | —            | —       |
|                                    | MMStar-Overall          | 1500      | 1500  | perception | 39.33    | —     | —   | —            | —     | —   | —        | —     | 38.73                                          | —            | —       |
|                                    | MMStar-Math             | 250       | 250   | reasoning  | 27.20    | —     | —   | —            | —     | —   | —        | —     | 27.60                                          | —            | —       |
|                                    | DynaMath-Average        | 5010      | 5010  | reasoning  | 17.78    | —     | —   | 19.26        | —     | —   | +1.48    | —     | 19.04                                          | —            | —       |
|                                    | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 2.20     | —     | —   | 2.40         | —     | —   | +0.20    | —     | 1.80                                           | —            | —       |
| LLaVA-OneVision-7B (Qwen2-Math-7B) | MathVista-Overall       | 1000      | 1000  | perception | —        | —     | —   | 60.00        | —     | —   | —        | —     | 62.50                                          | —            | —       |
|                                    | MathVista-General (NEW) | 460       | 460   | perception | —        | —     | —   | 59.31        | —     | —   | —        | —     | 61.53                                          | —            | —       |
|                                    | MathVista-Math (NEW)    | 540       | 540   | reasoning  | —        | —     | —   | 61.48        | —     | —   | —        | —     | 64.26                                          | —            | —       |
|                                    | MathVerse-Overall       | 788*      | 788*  | reasoning  | —        | —     | —   | 30.36        | —     | —   | —        | —     | 30.58                                          | —            | —       |
|                                    | MathVerse-T-D           | 788       | 788   | reasoning  | —        | —     | —   | 38.71        | —     | —   | —        | —     | 39.47                                          | —            | —       |
|                                    | MathVerse-T-L           | 788       | 788   | reasoning  | —        | —     | —   | 33.63        | —     | —   | —        | —     | 33.76                                          | —            | —       |
|                                    | MathVerse-V-I           | 788       | 788   | reasoning  | —        | —     | —   | 31.73        | —     | —   | —        | —     | 32.11                                          | —            | —       |
|                                    | MathVerse-V-D           | 788       | 788   | reasoning  | —        | —     | —   | 29.06        | —     | —   | —        | —     | 28.93                                          | —            | —       |
|                                    | MathVerse-V-O           | 788       | 788   | reasoning  | 17.64    | —     | —   | 18.65        | —     | —   | +1.01    | —     | 18.65                                          | —            | —       |
|                                    | MathVision              | 304       | 304   | reasoning  | 18.42    | —     | —   | 17.43        | —     | —   | -0.99    | —     | 15.79                                          | —            | —       |
|                                    | MMStar-Overall          | 1500      | 1500  | perception | 0.00     | —     | —   | 0.00         | —     | —   | +0.00    | —     | 0.00                                           | —            | —       |
|                                    | MMStar-Math             | 250       | 250   | reasoning  | 0.00     | —     | —   | 0.00         | —     | —   | +0.00    | —     | 0.00                                           | —            | —       |
|                                    | DynaMath-Average        | 5010      | 5010  | reasoning  | 19.66    | —     | —   | 20.72        | —     | —   | +1.06    | —     | 19.52                                          | —            | —       |
|                                    | DynaMath-Worst (NEW)    | 501       | 501   | reasoning  | 4.99     | —     | —   | 5.59         | —     | —   | +0.60    | —     | 4.79                                           | —            | —       |





## Blocked models — no on-disk data

These 4 models complete the 9-model enrichment-target set but have no step-25 score CSVs at all. Each is blocked by a different infrastructure issue, summarized here.

| Model              | Auto-routed math LLM        | Blocker                                                                                                                                                                                                                                                                                                                                                                                 | Status / handoff                                                                                                                                                                  |
|--------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LLaVA-OneVision-7B | Qwen2-Math-7B (HF)          | VLMEvalKit_brv venv missing the <i>llava</i> Python package. Eval jobs die silently with No module named 'llava'. <b>Same package issue affects LLaVA-1.5-7B today</b> — both will resume once the package is installed. Merge .pth already exists from earlier session.                                                                                                                | Handoff filed: notes/vlmevalkit_handoff_llava_next_install.md. One uv pip install command unblocks both llava-1.5 and llava-ov.                                                   |
| Qwen2.5-VL-7B      | Qwen2.5-Math-7B (HF)        | Two blockers. (1) Merge job crashed twice with an hf-xet Rust panic mid-download of Qwen2.5-Math-7B (range end out of bounds: 67M ≤ 34M). Worked around by pre-fetching via huggingface-cli download; merge re-submitted 2026-06-04 18:32. (2) Every eval job raises KeyError: 'Qwen2.5-VL-7B-Instruct' — VLMEvalKit's supported_VLM registry has the 3B and 2B but not the 7B variant. | Handoff filed: notes/vlmevalkit_handoff_qwen25vl_7b_registry.md (eval). Merge unblock used huggingface-cli download Qwen/Qwen2.5-Math-7B to bypass hf-xet.                        |
| InternVL2.5-8B     | internlm2-math-plus-7b (HF) | Merge job formerly killed by TERM_THREADLIMIT (fixed in run_pipeline.sh) and raised TypeError: dtype in merge_pmbt.py (fixed in code/merge_pmbt.py). <b>Merge .pth landed 2026-06-04 14:56 (15.5 GB)</b> . Eval still fails with 'InternLM2ForCausalLM' object has no attribute 'generate' — HF-cache modeling_internlm2.py doesn't inherit GenerationMixin.                            | Handoff filed: notes/internvl_handoff_generation_mixin.md. Merge side is unblocked by today's commits; eval side still needs the HF-cache patch (or transformers ≤ 4.49 pin).     |
| Qwen2.5-VL-3B      | — (none compatible)         | Hard-blocked by run_pipeline.sh (exit 1). Qwen2.5-VL-3B uses a 36-layer 2048-hidden Qwen2.5-3B-base LLM; Qwen2.5-Math ships only 1.5B (28L/1536H) and 7B (28L/3584H), so no Qwen2.5-Math variant is shape-compatible. Other math families (LLaMA-base, Mistral-base) are also architecturally incompatible with the Qwen2.5-3B backbone.                                                | Decision: <b>leave the block in place</b> . No shape-compatible math donor exists; lifting the block would just move the failure into merge_pmbt.py's tensor-shape mismatch path. |

## How close is local to the paper?

Counting only rows where both Local and Paper values exist:

| Slice                  | rows | mean  Gap | mean Gap | max  Gap | #  Gap >1 |
|------------------------|------|-----------|----------|----------|-----------|
| LLaVA — Baseline       | 13   | 0.92      | -0.71    | 3.56     | 6/13      |
| LLaVA — +Uniform 0.9   | 13   | 2.25      | -2.19    | 6.72     | 6/13      |
| Idefics — Baseline     | 13   | 0.68      | -0.51    | 2.70     | 2/13      |
| Idefics — +Uniform 0.9 | 13   | 1.08      | -0.97    | 2.00     | 7/13      |

## Observations

- **MathVista, MathVision, DynaMath:** Local tracks Paper tightly. All Baseline and most +Uniform cells are within ~0.5 pp; max |Gap| there is ~0.6 pp. The patched MathVista\_acc reproduces the BRV General/Math split to within ~1 pp.
- **MathVerse:** systematic shortfall. Local LLaVA baseline already runs 1–3 pp below paper; after merging with Dart-Prop the gap grows to 4–7 pp (Local +Uniform – Paper +Uniform). Idefics + MAMmoTH-1 shows a smaller but consistently negative MathVerse gap (1–2 pp).
- **Likely causes** for the MathVerse divergence: (1) MathVerse has a two-stage judge (extract + score) so it accumulates more gpt-4o-mini sampling variance than single-stage benchmarks; (2) the local LLaVA-Next checkpoint may differ slightly from the one BRV used; (3) the merge implementation may apply Dart-Prop slightly differently than BRV's official script.
- **Net interpretation:** the qualitative direction agrees with BRV (uniform-merge helps math; modest perception cost), but the absolute MathVerse numbers should be taken with the gap in mind. Re-running step 19 against the BRV-published gpt-4o-mini judge build, or re-pinning the LLaVA checkpoint, would close most of the gap.

## Coverage gaps in the local data

- **InternVL2-76B** — no local eval (size makes it expensive); BRV's InternVL row can only come from the paper.
- **MathVerse-Overall** — the local pipeline doesn't precompute this row, but it is reconstructed in this PDF as the simple mean of the five mode scores (matching BRV).
- **DynaMath Worst-Case for the paper column** — BRV Tables 2/3 do not report it (paper only reports Average).
- **Older results/13-weight-merge/\* JSON outputs** — different format with no cached judge xlsx, skipped in the re-score.

## Provenance

- Local score CSVs read from results/25-merge/<model>/<tv>/eval\_baseline and .../eval\_uniform\_0.9.
- Paper values: BRV (Chen et al., 2025) Table 2 (LLaVA, +Dart-Prop column) and Table 3 (Idefics, +MAMmoTH-1 column).

- MathVista General/Math rows populated by the patched VLMEvalKit\_brv/vlmeval/dataset/utils/mathvista.py:MathVista\_acc.
- DynaMath worst-case row already produced by VLMEvalKit\_brv/vlmeval/dataset/dynamath.py:226-244; summarizer (code/summarize\_eval\_results.py) patched to read Setting=Worst Case.
- Re-scoring run: 406 files in ~107 s on the modern\_vlms venv; no judge re-calls.